

XRPL Validator — Daily Engineering Report • 2026-09-27

Generated 2026-09-28 00:21 EDT. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the live native-engine shadow, the reference rippled node, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **3 commits** (xrpl-validator 3)
- **3 engine findings fixed** (F403–F405)
- Live shadow: 15,098/15,099 ledgers fully matched, 2,260,222 transactions, 1 engine receipts in the current soak
- Amendment **PermissionDelegationV1_1** held majority (since 2026-09-24 21:25 UTC); activation due ~2026-10-08 21:25 UTC
- Amendment **fixBatchV1_2** held majority (since 2026-09-25 14:12 UTC); activation due ~2026-10-09 14:12 UTC
- Amendment **BatchV1_1** held majority (since 2026-09-25 14:46 UTC); activation due ~2026-10-09 14:46 UTC

Live validator and native engine

Snapshot at generation time.

Measure	Value
Ledgers compared (current run)	15,099
Ledgers fully matched	15,098
Transactions applied natively	2,260,222
Result mismatches	1
Byte mismatches	1
Missing / extra keys	0 / 6
Batch inner-result mismatches	0
Engine receipts (current soak)	1
Live engine build running since	2026-09-27 08:00 EDT
Reference rippled	3.4.1, full, validated #107,285,084

Engine findings fixed

- **fix(xrpl-ledger): F403 — an IOU Clawback moves the exact amount through rippleCredit** (e7d7655 , xrpl-validator) Mainnet #107263299 DC750F7C1704: an issuer clawed 200,000 XXPAYMENTS back from a line holding 200,000.00000000005, leaving 0.00000000005. rippled's Clawback (applyHelper) moves min(accountHolds, Amount) from the holder to the issuer with directSendNoFee: exact decimal amounts and rippleCredit's line mechanics. The engine parsed both amounts as f64, subtracted, and wrote 0.00000000049476... — f64 cannot hold ...

- **fix(xrpl-ledger): F404 — a ticket-funded XRP Payment's reserve excludes the spent ticket** (ae84a26 , xrpl-validator) Mainnet #107266300 F31123B49AFE: a 5,877-drop XRP Payment paid for with a Ticket, from an account holding 3.002177 XRP at OwnerCount 10 (ten tickets). rippled's Transactor::apply deletes the ticket being spent (consumeSeqProxy: OwnerCount 10 -> 9) before Payment::doApply's funding guard, mPriorBalance < amount + max(reserve, fee): the reserve is $1 + 0.2 \times 9 = 2.8$ XRP and mainnet succeeded. The engine makes that ...
- **fix(xrpl-ledger): F405 — an AMM withdrawal by the issuer checks the reserve for its own currency** (51b0cbe , xrpl-validator) Mainnet #107278773 784FE06097CB: a tfWithdrawAll from an XRP/001 pool by rJENNivB, the account that issues 001, holding 1.41 XRP at OwnerCount 2. rippled's AMMWithdraw sufficientReserve (fixAMMv1_2) asks only whether keylet::trustLine(account, issue) exists, and an issuer never has a line to itself, so the check runs for the issuer's own currency too: a third object needs $1 + 0.2 \times 3 = 1.6$ XRP, and with the pool's ...

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **Soak #32: two engine receipts, both fixed.** Soak #32 of build c79ee89 (from 13:21 EDT on 2026-09-26) recorded two engine receipts, F403 and F404 below; it ended with this morning's deployment at 08:01 EDT. Over the soak the shadow compared 16,823 ledgers and the state-hash check matched 16,828, with no state-hash mismatch: the receipts are differences in the native engine's own results, found by its comparison against libxrpl, not in the validator's state.
- **Finding F403 (Clawback of an issued token).** At ledger 107,263,299 an issuer clawed back 200,000 units of its token from a trust line holding 200,000.0000000005. rippled moves the smaller of the holder's balance and the amount with exact decimal arithmetic and left 0.0000000005 on the line; the engine did the subtraction in binary floating point, which cannot represent 200,000.0000000005, and wrote about 0.00000000049476. Clawback now moves the amount through the engine's exact trust-line credit, the one payments use, which also brings rippled's line bookkeeping. The mainnet transaction is pinned as a regression vector. Fixed in e7d7655.
- **Finding F404 (XRP payment paid for with a Ticket).** At ledger 107,266,300 an account holding 3.002177 XRP with ten tickets (owner count 10) sent 5,877 drops using one of them. rippled deletes the ticket being spent before it checks the payment against the reserve, so the reserve was $1 + 0.2 \times 9 = 2.8$ XRP and mainnet accepted the payment; the engine checked earlier, still counting the ticket (3.0 XRP), and refused it. The same pattern as F402 (TicketCreate, 2026-09-25). Fixed in ae84a26. Two other transaction types, EscrowCreate and TrustSet, also check their reserve after the ticket is consumed in rippled while the engine checks before; no mainnet transaction has exercised the difference, and devnet transactions will be generated to test both.
- **Deployment.** Build ae84a26 replaced the live build at 08:01 EDT, after its gate (1,691 suite tests and 745 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches), by warm restart: it rebuilt its state in 22.4 seconds and resumed two ledgers behind the network. The health check matched 410 ledgers with no mismatches, and soak #33 began at 08:28 EDT (seed ledger 107,269,979). Besides F403 and F404 it adds two read-only status endpoints for the operations monitor (the validator's connections, and its amendment votes) and changes the votes.
- **Amendment votes.** The validator now votes for exactly the pending amendments its engine implements and has verified: PermissionDelegationV1_1 is added (implemented and verified against devnet), XChainBridge and

fixXChainRewardRounding are removed (the engine's cross-chain bridge code has never run against a real bridge). 45 votes instead of 46. The first voting ledger after the deployment, 107,270,143, carried the 45 votes and reached the reference node.

- **Amendment readiness.** The operations monitor now warns, with the date, when an amendment with majority on mainnet is one the engine is not ready for. Today that is fixBatchV1_2, due about 2026-10-09 14:12 UTC; its source is still unpublished.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

Amendments enabled on mainnet: 94.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **On-Ledger Asynchronous Multisig Proposals** (custody) Add a PendingTxn ledger object (ltPENDING_TXN , owner-reserved by the multisig account, keylet::pendingTxn(account, sfSequence)) that holds one s
- **Minimum-Fill Offers to Stop Dust Execution** (dex) Add an optional sfMinFillAmount (STAmount, in the offer's TakerGets currency) to OfferCreate and the Offer SLE. In BookStep, a crossing that w
- **Tranche Vesting Escrows in One Object** (payments) Extend EscrowCreate and the Escrow SLE with three optional fields: sfVestingInterval (UInt32 seconds), sfVestingPeriods (UInt8, 2–48), and `sf

Appendix — commits

Time	Repo	Commit	Subject
01:29	xrpl-validator	e7d7655	fix(xrpl-ledger): F403 — an IOU Clawback moves the exact amount through rippleCredit
04:12	xrpl-validator	ae84a26	fix(xrpl-ledger): F404 — a ticket-funded XRP Payment's reserve excludes the spent ticket
17:42	xrpl-validator	51b0cbe	fix(xrpl-ledger): F405 — an AMM withdrawal by the issuer checks the reserve for its own currency